

MATHEMATICS XII

Questions Distribution

According to the Chapters.

This Distribution is carried out According to
NEW BOOK OF MATHEMATICS.

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QUESTIONS		Ex. No.	YEARS
Unit: 1	Introduction to Symbolic Package MAPLE		
Section B			
Section C			

Unit: 2	Functions and Limits		
Section B			
2. i) If $f(x) = \frac{x+1}{3}$ and $g(x) = 3x + 5$ are two functions then show that $(fog)^{-1} = g^{-1}of^{-1}$.	Rev. Ex.	2025 P.E.	
ii) Evaluate any one of the following: $*\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{\sin^2 x}$ $*\lim_{x \rightarrow e} \frac{\ln x - 1}{x - e}$	2.3		
2. i) Find the value of 'p' if $f(x) = \begin{cases} \frac{\sin px}{x} & ; x \neq 0 \\ 2 & ; x = 0 \end{cases}$ is continuous at $x = 0$. (Same question with different variable)	2.4	2025 S. G.	
ii) Evaluate any one of the following: $*\lim_{x \rightarrow 0} \frac{\sin \sqrt{a}x}{x}$ $*\lim_{x \rightarrow 0} \frac{\sqrt{x+3} - \sqrt{3}}{x}$	2.3		
2. i) Evaluate any one of the following:	2.3	2024	

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(a) $\lim_{x \rightarrow 0} \frac{\sin x}{x}$	(b) $\lim_{x \rightarrow 0} \frac{\ln x - 1}{x - e}$		P.E.
ii) Test continuity and discontinuity of the following function: $f(x) = \begin{cases} 2 + x, & \text{when } x < 3 \\ 5 - 2x, & \text{when } x \geq 3 \end{cases}$ at $x = 3$		2.4	
EXAMPLE (II), TOPIC 2.8.3			
2. i) Evaluate any one of the following: a) $\lim_{n \rightarrow 0} \frac{\sqrt{x+3} - \sqrt{3}}{x}$ b) $\lim_{n \rightarrow 0} \frac{1 - \cos nx}{1 - \cos nx}$		2.3	2024 S. G.
ii) Find the value of 'k' if: $f(x) = \begin{cases} \frac{\sin kx}{x} & : x \neq 0 \\ k & : x = 0 \end{cases}$ is continuous at $x = 0$		2.4	
Section C			

Unit: 3	DIFFERENTIATION		
Section B			
iii) Find the derivative of $f(x) = \sin x$ by ab-initio/first principle method.	3.4	2025 P.E.	
TOPIC 3.5.1			
iii) Find the derivative of the function $f(x) = 3 - x^3$ by the initio first principle.	3.2	2025 S. G.	
iii) If $y = \sqrt{\sin x} \sqrt{\sin x} \sqrt{\sin x} \dots$, show that $(2y - 1) \frac{dy}{dx} = \cos x$ NOTE: The above question in the board's Paper has some typo-math error. Correct question is given below: If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots}}}$, show that $(2y - 1) \frac{dy}{dx} = \cos x$		2024 P.E.	Rev. EX.
iv) Differentiate any one of the following with respect to x. (a) $y = \tan^{-1} \left(\frac{\sin 2x}{1 + \cos 2x} \right)$	3.4		
(b) $y = \tan h^{-1}(\tan x^3)$	3.6		

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iii) Find the derivative of $f(x) = \sec \sqrt{x}$ by using first principle method.	3.4	2024 S. G.
v) Find $\frac{dy}{dx} : y^3 - 3y + 2x = 0$ OR $y - xy + \sin y = 0$	3.4	
Section C		
4. Differentiate any two of the following: $*y = \tan^{-1} \left(\frac{\sin 2x}{1 + \cos 2x} \right)$	3.4	2025 P.E.
$*y = x \cosh^{-1} x - \sqrt{x^2 - 1}$	3.6	
$*x\sqrt{1+y} + y\sqrt{1+x} = 0$	3.4	
4. Differentiate any two of the following: $*y = \tan(e^{\sin h^{-1} x})$	3.6	2025 S. G.
$*y = \sin^{-1} \sqrt{\frac{1 - \cos x}{2}}$	3.4	
$*y = \cos^2 x$ w.r.t $\sin^2 x$	3.4	
EXAMPLE 2(ii): TOPIC 3.5.1		

Unit: 4	HIGHER ORDER DERIVATIVES AND APPLICATIONS		
Section B			
iv) Obtain first four terms of Taylor's series for $f(x) = \ln x$ centered at $a = 1$.	4.2	2025 P.E.	
iv) Obtain the first four terms of the Taylor's series for $f(x) = e^x$ centered $x = 1$.	4.2	2025 S. G.	
v) Find the maximum and minimum value(s) of $f(x) = x - 2x + 3$.	4.4	2024 P.E.	
iv) Show that $f(x) = \tan^2 x$ is decreasing at $x = \frac{3\pi}{4}$	4.4	2024 S. G.	
Section C			
5. Find the maximum and minimum value of the $f(x) = x^3 - 9x^2 + 15x + 3$	4.4	2025 P.E.	
5. Find the right-angled triangle of maximum area with a hypotenuse of length h .	4.4	2025 S. G.	

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4. Find the first four terms of Taylor's Series of $f(x) = \ln(1+x)$ at $b = 2$.	4.2	2024 P.E.
EXAMPLE (vi), TOPIC 4.2.1		
4. Find the first four terms of Taylor's Series of $f(x) = \cos x$ at $a = \frac{\pi}{4}$.	4.2	2024 S.G.
EXAMPLE (ii), TOPIC 4.2.1		

Unit: 5 DIFFERENTIATION OF VECTOR FUNCTIONS

Section B

v) Show that the function $\vec{r}(t) = \sin^2 t \hat{i} + \tan t \hat{j} + \frac{1}{t} \hat{k}$ is continuous at $t = \frac{\pi}{4}$.	5.1	2025 P.E.
v) Show that the function $\vec{r}(t) = t \hat{i} + \frac{1}{t+1} \hat{j}$ is continuous at $t = 0$.	5.1	2025 S.G.
EXAMPLE :2, TOPIC 5.2.2		
vi) A particle moves along a curve whose parametric equations are $x = e^{-t}$, $y = 2\cos 3t$ and $z = 2\sin 3t$. Find the magnitude of the acceleration at $t = 0$.	5.2	2024 P.E.
EXAMPLE :1, TOPIC 5.4.2		
vi) A particle moves so that its position as a function of time is given by $\vec{r}(t) = t\hat{i} + 4t^2\hat{j} + t\hat{k}$. Write expressions for its: a) Velocity b) acceleration as functions of time	5.2	2024 S.G.

Section C

Unit: 6 INTEGRATION

Section B

vi) Compute the definite integral $\int_{\frac{\pi}{2}}^{\pi} \cos x \, dx$ by using basic properties.	6.6	2025 P.F.
EXAMPLE :6, TOPIC 6.6.3		
vii) Find the area, above x-axis under the curve $y = 3x^2 + 2$ between the ordinates $x = 1$ and $x = 2$.	6.7	
vi) Compute the definite integral $\int_{-2}^2 (x^4 + 2x^2) \, dx$	6.6	2025

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by using basic properties.		S.G.
vii) Find the area, above x-axis under the curve $y = 3\sec^2 x$ between the ordinates $x = \frac{\pi}{6}$ and $x = \frac{\pi}{3}$.	6.7	
vii) Compute the following definite integral by using basic properties. $\int_{-2}^2 (x^4 + 2x^2) \, dx$	6.6	2024 P.E.
viii) Find the area, above the x-axis under the curve $y = x^2 + 1$ between the given ordinates $x = 2, x = 4$.	6.7	
EXAMPLE :1, TOPIC 6.6.5		
vii) Compute the definite integral by using basic properties: $\int_{-\pi/2}^{\pi/2} \sin^2 x \cos^2 x \, dx$	6.6	2024 S.G.
viii) Find the Area, above the x-axis under the curve $y = 3x^2 + 2$, between the ordinates $x = 1, x = 2$.	6.7	

Section C

3. Evaluate any two of the following: * $\int \frac{(5x-2)}{(x-3)(x+7)} \, dx$ (by using partial fraction)	6.5	2025 P.E.
* $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} x \sin 2x \, dx$ (by using integration by parts)	6.6	
* $\int x^2 \sqrt{3-x^2} \, dx$ (by using trigonometric substitution)	6.6	
3. Evaluate any two of the following: * $\int 2x \sec^{-1} x \, dx$ (by using integration by parts)	6.4	2025 S.G.
* $\int \frac{\cos x \, dx}{\sin x (2 + \sin x)}$ (by using partial fraction)	6.5	
EXAMPLE :2(ii)		
* $\int_2^{2\sqrt{3}} \frac{x^3 \, dx}{\sqrt{x^3+4}}$ (by using trigonometric substitution)	6.6	
3. Evaluate any two of the following: (a) $\int \cos^2 x \ln^3 x \, dx$ EXAMPLE :3, TOPIC 6.3.2	6.2	2024 P.E.
(b) $\int_{-6}^{-2\sqrt{3}} \frac{dx}{x\sqrt{x^2-9}}$ EXAMPLE :8, TOPIC 6.6.3	6.6	
(c) $\int \cos(\ln x) \, dx$	6.4	

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9. Solve the differential equation: $(x^2 + y^2)dx = 2xy dy$ EXAMPLE:1, TOPIC 10.3.1,	10.2	2025 P.E.
9. —OR—The population of a certain town is directly proportional to the square root of the present population at any time initially is 20,000. How much the population after 10 years and after how much time the population be doubled.	10.2	
9. Solve the differential equation: $(x - y - 1)dx - (x - y - 5)dy = 0$ EXAMPLE : TOPIC 10.3.1, Page 362	10.2	2025 S. G.
9. —OR—Solve the problem by differential equation, a ball is thrown upward vertically with velocity 49 m/s and $g = -9.8 \text{ m/s}^2$. Find: i) time when the body at maximum height, ii) height with $t = 3$ seconds iii) maximum height. EXAMPLE : TOPIC 10.3.1, Page 361	10.2	
8. Solve the differential equation $\frac{dy}{dx} = \frac{x-2y+2}{2x+y-1}$ EXAMPLE : TOPIC 10.3.1, Page 361	10.2	2024 P.E.
8. Solve the differential equation $(x^2 + y^2)dx = 2xy dy.$	10.2	2024 S. G.

Unit: 11	PARTIAL DIFFERENTIATION	
	Section B	
xiv) Verify Euler's theorem for the homogenous function $f(x, y) = \cos\left(\frac{x}{y}\right)$.	11	2025 P.E.
xiv) The volume of the cone is given by formula $V = \frac{1}{3}\pi r^2 h$. Differentiate V w.r.t their independent variable. (where r and h are its independent variables).	11	2025 S. G.
9. Given that $u = \sec^{-1}\left(\frac{x^2 - y^2}{x + y}\right)$ show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \cot u.$	REV. EX.	2024 P.E.
xii) The volume of the cone is given by the formula	11	2024

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$V = \frac{1}{3}\pi r^2 h$. Differentiate V with respect to their independent variables.		S. G.
Section C		
9. If $u = x^2(y - x) + y^2(x - y)$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = -2(x - y)^2$.	11	2024 S. G.
Unit: 12	INTRODUCTION TO NUMERICAL METHODS	
	Section B	
	Section C	
10. Use Regula Falsi method to find a root of the equation $x^3 - 2x - 5 = 0, [2, 3]$ upto three iterations.	12.2	2024 P.E.
10.—OR—Evaluate $\int_0^1 e^x dx$ by Simpson's $\frac{1}{3}rd$ rule with $n = 6$. EXAMPLE :1, TOPIC 12.2.1	12.4	
10. Use Newton Raphson method to find the real root of $f(x) = x^3 - x - 1$ upto three iterations with initial guess $x_0 = 1$.	12.3	2024 S. G.
Unit:	MISCELLANEOUS QUESTION	
	Section B	
vi) A particle moves along a curve whose parametric equations are $x = e^{-t}$, $y = 2\cos 3t$ and $z = 2\sin 3t$. Find the magnitude of the acceleration at $t = 0$.	4.4	2024 P.E.

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7. Find the equation of tangent(s) to the circle $x^2 + y^2 = 25$ which is perpendicular to $3x + 4y + 1 = 0$.	8.3	2025 S. G.
6. Prove that two circles $x^2 + y^2 + 2gx + c = 0$ and $x^2 + y^2 + 2fx + c = 0$ touch each other, if $\frac{1}{f^2} + \frac{1}{g^2} = \frac{1}{c}$.	8.3	2024 P.E.
6. Find the equation of tangents to the circle $x^2 + y^2 = 25$ which is: a) Parallel to $3x + 4y + 1 = 0$ b) Perpendicular to $3x + 4y + 1 = 0$.	8.3	2024 S. G.

Unit: 9

PARABOLA, ELLIPSE AND HYPERBOLA

Section B

xi) Find the equation of parabola whose vertex and focus are $(0, 0)$ and $(0, -2)$ respectively.	9.1	2025 P.E.
xii) Find the eccentricity of ellipse if axes are 32 and 24 respectively.	9.3	
xi) Find the equation of parabola whose focus is $(2, 4)$ and equation of directrix is $x + 3 = 0$. EXAMPLE : TOPIC 9.3.2	9.1	2025 S. G.
xii) Find the distance between the vertices of ellipse $9x^2 + 13y^2 = 117$.	9.3	
xii) Find the equation of the parabola whose focus $(3, 4)$ and directrix is the line $x + y - 1 = 0$.	9.1	2024 P.E.
xiii) If e_1 and e_2 are eccentricities of two conjugate hyperbolas then prove the relation $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$. Similar as Question No.7	9.6	
xv) Find the eccentricity of an ellipse if length of latus rectum is equal to half of its major axis.	9.3	
xiii) Determine vertex, focus and length of latus rectum of the parabola $y^2 - 6y + 8x - 23 = 0$.	9.1	2024 S. G.
xiv) If Ellipse $9x^2 + 13y^2 = 117$ then find: a) Distance between vertices b) distance between foci.	9.3	

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xiv) —OR— Show that the eccentricities e_1 and e_2 of two conjugate hyperbola satisfy the relation $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$ Similar as Question No.7	9.6	
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Section C

8. Find foci, eccentricity, vertices and latus rectum of hyperbola $\frac{(x-3)^2}{64} - \frac{(y+4)^2}{36} = 1$ EXAMPLE :3, TOPIC 9.10.2	9.5	2025 P.E.
8. Find foci, eccentricity, vertices and equation of directrices of the hyperbola $16(y+2)^2 - 9(x-2)^2 = 144$. EXAMPLE :2, TOPIC 9.10.3	9.5	2025 S. G.
7. Find the eccentricity, foci, vertices and directrices of hyperbola $9x^2 - 16y^2 - 144 = 0$. EXAMPLE :1, TOPIC 9.10.3	9.5	2024 P.E.
7. Find the centre, foci, vertices and length of latus rectum of the hyperbola $9x^2 - 16y^2 - 144 = 0$. EXAMPLE :1, TOPIC 9.10.3	9.5	2024 S. G.

Unit: 10

DIFFERENTIAL EQUATIONS

Section B

xiii) Show that $y = Ae^{2x} + Be^{3x}$ is the general solution of $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$.	10.1	2025 P.E.
xiii) Obtain the differential equation by eliminating arbitrary constant from the relation $y = C_1e^x + C_2e^{-x}$. EXAMPLE :2(b), TOPIC 10.2.1	10.1	2025 S. G.
xiv) Find the orthogonal trajectory of the family of parabola $y^2 = 4ax$.	10.3	2024 P.E.
xv) Find the orthogonal trajectory of the curve $y = ax^2$.	10.3	2024 S. G.

Section C

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(d) $\int \frac{\sec^2 x dx}{(1 + \tan x)(2 + \tan x)}$	6.5	2024 S. G.
3. Evaluate any two of the following:	6.3	
(a) $\int \frac{\sqrt{x^2-25}}{x} dx$ EXAMPLE :1(III), TOPIC 6.3.3	6.4	
(b) $\int e^{ax} \cos bx dx$ Similar as Question No.3(x)	6.5	
(c) $\int \frac{\sec^2 x dx}{(1 + \tan x)(2 + \tan x)}$	6.2	
(d) $\int \frac{1+2x}{1-x} dx$		

Unit: 7 PLANE ANALYTIC GEOMETRY: STRAIGHT LINE

Section B

viii) Find the equation of the line which passes through (5,6) and y-intercept is twice that of x-intercept.	7.3	2025 P.E.
ix) If A(2,5), B(3,7) and C(0,8) are the vertices of a triangle then find equation of median through A.	7.6	
viii) Find the equation of the line which passes through (11, -5) and parallel to a line having slope $\frac{3}{2}$.	7.3	2025 S. G.
ix) If A(2,5), B(3,7) and C(0,8) are the vertices of a triangle then find equation of right bisectors of side AC.	7.6	
ix) Z(4,5) and X(7,-1) are two given points and the point Y divides the line segment ZX externally in the ratio 4:3. Find the coordinate of Y.	7.1	2024 P.E.
x) Show that the lines $2x - 3y + 5 = 0$, $3x + 4y - 7 = 0$ and $9x - 5y + 8 = 0$ are concurrent. EXAMPLE :1, TOPIC 7.7.1	7.6	
ix) The line segment P(-8,10) and Q(6,-4) is cut by x and y axes at A and B respectively. Find the ratios in which A and B divides PQ.	7.1	2024 S. G.
x) The line through (2,5) and (-3,-2) is perpendicular to the line through (4,-1) and (x,3), find x.	7.2	

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Section C

6. If a point P(k,7) divides the line segment joining A(8,9) and B(1,2) in a ratio m:n then find ratio m:n and value of k.	7.1	2025 P.E.
6. Find the equation of a line through the intersection of the lines $3x - 4y + 1 = 0$, $5x + y - 1 = 0$ and cutting off equal intercepts from the axes.	7.5	2025 S. G.
5. Find the angle between the lines $x^2 - 5xy + 6y^2 = 0$.	7.8	2024 P.E.
5. —OR— Find the equation of a line through the intersection of the lines $5x - 6y = 1$, $3x + 2y + 5 = 0$ and perpendicular to $5y - 3x = 11$.	7.5	
5. Find the equation of a line through the intersection of lines $2x + 3y + 1 = 0$ and $3x - 4y = 5$ and passing through the point (2,1).	7.5	2024 S. G.
5. —OR— The gradient of one of the lines $ax^2 + 2hxy + by^2 = 0$ is twice that of the other show that $8h^2 = 9ab$.	7.8	

Unit: 8 CIRCLE

Section B

x) Find the equation of circle containing the points (0,0), (0,3) and (-4,0).	8.2	2025 P.E.
x) Find the equation of circle containing the points (1,-2) and (3,-4) and touching x-axis.	8.2	2025 S. G.
xi) Find the equation of circle containing the points (1,-2) and (3,4) and touching x - axis.	8.2	2024 P.E.
(xi) OR— Find the equation of circle passing through (-3,-4) and is concentric with the circle $x^2 + y^2 - 6x + 8y - 24 = 0$.	8.1	
xi) Find the equation of circle passing through two points (1,0) and (0,1) and touches the line $x + y = 0$. EXAMPLE :1, TOPIC 8.4.1 (d)	8.2	2024 S. G.
Section C		
7. Find the condition that the line $3x + 4y = c$ may touch the circle $x^2 + y^2 = 8x$.	8.3	2025